WORKED SOLUTION

STEP III 2013, Question 11

Mechanics • Elastic Springs and Energy

STEP III 2013, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s3-q11

The key idea

Everything is fixed by symmetry: the triangle stays horizontal, its centre stays directly below the suspension point, and it can only move up and down. So one particle and one spring tell the whole story. The geometry of the vertical plane through a spring gives its length in terms of θ. Resolving vertically gives the equilibrium, and conservation of energy (elastic, gravitational and kinetic) gives the speed.

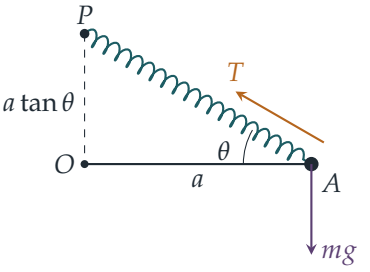
Solution

Model and geometry

Treat the masses as particles, the rods as light and rigid, and the springs as light and obeying Hooke's law. Let P be the point of suspension and O the centre of the triangle, directly below P. The distance from the centre of an equilateral triangle of side s to a vertex is $s/\sqrt{3}$, so each vertex is at horizontal distance

$$OA = \frac{\sqrt{3}a}{\sqrt{3}} = a$$

from O. Take a vertex A and look at the vertical plane through P, O and A:



The spring PA makes angle θ with the horizontal OA, so the spring has length $a/\cos\theta$ and A is a depth $a\tan\theta$ below P.

The tension

The extension of each spring is

$$x = \frac{a}{\cos\theta} - a = \frac{a(1-\cos\theta)}{\cos\theta},$$

so by Hooke's law, with modulus kmg and natural length a,

$$T = \frac{kmg}{a}\,x = \frac{kmg(1-\cos\theta)}{\cos\theta},$$

as required.

Equilibrium at $\theta = \frac{1}{6}\pi$

The forces on the particle at A are its weight, the spring tension along AP and the forces from the two rods. The rods lie in the horizontal plane, so their forces are horizontal.

Ask yourself Why can we ignore the rod forces?

We resolve vertically, and the rod forces have no vertical component. (Horizontally the tension component $T\cos\theta$ is balanced by the rods, which is why the rods are needed, but we don't need that equation.) Resolving vertically at A:

$$T\sin\theta = mg \quad\Longrightarrow\quad k(1-\cos\theta)\tan\theta = 1.$$

At $\theta = \frac{1}{6}\pi$, $\cos\theta = \frac{\sqrt{3}}{2}$ and $\tan\theta = \frac{1}{\sqrt{3}}$, so

$$k\cdot\frac{2-\sqrt{3}}{2}\cdot\frac{1}{\sqrt{3}} = 1 \quad\Longrightarrow\quad k = \frac{2\sqrt{3}}{2-\sqrt{3}} = \frac{2\sqrt{3}(2+\sqrt{3})}{(2-\sqrt{3})(2+\sqrt{3})} = 4\sqrt{3}+6,$$

since $(2-\sqrt{3})(2+\sqrt{3}) = 1$. This is the required value.

The speed at the equilibrium position

By symmetry the triangle stays horizontal with its centre below P, so all three particles move vertically with the same speed. No external force other than gravity and the springs does work, so energy is conserved. The elastic energy in a spring of extension x is $kmg\,x^2/(2a)$.

Start ($\theta = \frac{1}{3}\pi$, at rest). Here $\cos\theta = \frac{1}{2}$, so the extension is $x_1 = a$, and the depth below P is $a\tan\frac{1}{3}\pi = \sqrt{3}a$.

Equilibrium ($\theta = \frac{1}{6}\pi$). The extension is

$$x_2 = a\left(\frac{2}{\sqrt{3}} - 1\right) = \frac{a(2-\sqrt{3})}{\sqrt{3}}, \qquad x_2^2 = \frac{a^2(7-4\sqrt{3})}{3},$$

and the depth is $a\tan\frac{1}{6}\pi = \frac{a}{\sqrt{3}}$. So each particle rises by

$$\sqrt{3}a - \frac{a}{\sqrt{3}} = \frac{2a}{\sqrt{3}}.$$

Elastic energy released by the three springs:

$$3 \cdot \frac{kmg}{2a}\left(x_1^2 - x_2^2\right) = \frac{3kmg}{2a} \cdot \frac{a^2(3 - 7 + 4\sqrt{3})}{3}$$
$$= 2kmga(\sqrt{3} - 1).$$

With $k = 4\sqrt{3} + 6$, $k(\sqrt{3} - 1) = 12 - 4\sqrt{3} + 6\sqrt{3} - 6 = 6 + 2\sqrt{3}$, so the elastic energy released is $4mga(3 + \sqrt{3})$.

Gravitational energy gained by the three particles: $3mg \cdot \dfrac{2a}{\sqrt{3}} = 2\sqrt{3}\,mga$.

The difference becomes kinetic energy of the three particles, each moving with speed V:

$$\tfrac{3}{2}mV^2 = 4mga(3 + \sqrt{3}) - 2\sqrt{3}\,mga = mga\left(12 + 2\sqrt{3}\right),$$
$$V^2 = \frac{2ag}{3}\left(12 + 2\sqrt{3}\right) = \frac{4ag}{3}\left(6 + \sqrt{3}\right),$$

as required.

Checks

- Signs: at $\theta = \frac{1}{3}\pi$ the vertical pull of each spring is $T\sin\theta = kmg \cdot \frac{\sqrt{3}}{2} \approx 11.2\,mg > mg$, so the triangle does start by moving *up* towards the equilibrium, as the energy equation assumes.
- The answer is positive and has the units of a speed squared (ag).
- Taking $a = 1.3\,\text{m}$ and $g = 9.81\,\text{m s}^{-2}$, stepping the equation of motion $\ddot{y} = g - \frac{T}{m}\sin\theta$ forward numerically from rest at $\theta = \frac{1}{3}\pi$ gives $V^2 \approx 131.48\,\text{m}^2\,\text{s}^{-2}$ at $\theta = \frac{1}{6}\pi$, matching $\frac{4ag}{3}(6 + \sqrt{3})$.

Common slips

- Taking the distance from the centre to a vertex to be $\sqrt{3}a$ or $\frac{\sqrt{3}}{2}a$ instead of a.
- Forgetting that there are three springs and three particles: the factor 3 appears in the elastic, gravitational and kinetic terms.
- Getting the direction of motion wrong: the particles *rise* between the two positions, so gravitational energy is gained, not lost.

Beyond STEP

The triangle has only one degree of freedom here because symmetry rules out every other motion. Choosing a single coordinate (here θ, or the depth y) and writing the energy in terms of it is the starting point of Lagrangian mechanics. Expanding the

potential energy about the equilibrium would also give the period of small vertical oscillations. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops energy methods and small oscillations in this spirit.

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